

Quadratic Maximum Likelihood power-spectrum recovery on GLASS mock weak-lensing catalogues: sky-coverage and depth requirements, and implications for TR1

Weak Lensing QML pipeline

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Abstract

We test a C++ implementation of the Quadratic Maximum Likelihood (QML) power-spectrum estimator against the pseudo- C_ℓ (PCl, via **NaMaster**) and an independent catalogue-based (**heracles**) estimator, using full-sky log-normal weak-lensing mock catalogues generated with **GLASS**. We scan a 3×3 grid of galaxy number density ($n = 1, 10, 20 \text{ arcmin}^{-2}$) and survey footprint (three Galactic+ecliptic cuts giving $f_{\text{sky}} \approx 15\%, 30\%, 55\%$) at two map resolutions, $N_{\text{side}} = 64$ and $N_{\text{side}} = 128$. Consistent with Maraio, Hall & Taylor (2022), we find that the QML estimator recovers a B -mode power spectrum consistent with pure shot noise, and does so with smaller variance than PCl, but only once *both* the sky coverage and the galaxy density are sufficiently large; with a small, disconnected footprint the recovered B -mode is dominated by mask-induced E -to- B leakage regardless of depth. We then examine the real TR1 Euclid-like shear catalogue and find its effective sky coverage ($f_{\text{sky}} = 0.371\%$ at $N_{\text{side}} = 256$) is far below what our simulations show is needed for a clean QML null test, while its per-tomographic-bin galaxy density ($\approx 2.8\text{--}3.2 \text{ arcmin}^{-2}$) is in the regime our grid identifies as usable *if* coverage were adequate. This suggests that a per-bin QML analysis of a TR1-like survey becomes feasible mainly once the footprint is enlarged, and that our $N_{\text{side}} = 128$, low-density grid points are a reasonable proxy for that regime.

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1 Introduction

The Quadratic Maximum Likelihood (QML) estimator provides a statistically optimal (minimum-variance, unbiased) estimate of the angular power spectrum of a masked, noisy Gaussian random field, at a computational cost that is markedly higher than the pseudo- C_ℓ (PCl) approach commonly used in current weak-lensing analyses. Maraio, Hall & Taylor (2022) (arXiv:2207.10412) present a new, computationally efficient QML implementation targeting Stage-IV weak-lensing surveys and demonstrate, among other things, that (i) QML significantly outperforms PCl in variance for B -mode null tests and for low- ℓ E -modes, and (ii) this remains true when the underlying field is a non-Gaussian (log-normal) realisation rather than a Gaussian one, even though the QML estimator’s optimality is formally derived under a Gaussian likelihood assumption.

This document does *not* re-derive the QML formalism, the Fisher-matrix estimator, or its optimality properties – these are covered in detail in Maraio, Hall & Taylor (2022). Instead, we describe (Section 2) the mock weak-lensing catalogues we generated with GLASS to exercise our own C++ QML implementation, note why using log-normal GLASS fields as an input is justified by Maraio, Hall & Taylor (2022)’s Section 4.5 (Section 3), present the resulting $3 \times 3 \times 2$ grid of sky-coverage/depth/resolution comparisons between QML, PCl and **heracles** (Section 4), and finally use the real TR1 Euclid-like catalogue (Section 5) to argue what sky coverage and depth would be needed to apply this pipeline, per tomographic bin, to real Stage-IV-like data.

2 GLASS simulation setup

All catalogues are generated with GLASS (Generator for Large Scale Structure), which produces full-sky, shell-by-shell log-normal matter-density and convergence/shear fields from an input angular matter power spectrum, and Poisson-samples galaxy positions and ellipticities from those fields. Angular matter power spectra are computed with CAMB for a flat Λ CDM fiducial cosmology,

$$h = 0.7, \quad \Omega_c = 0.25, \quad \Omega_b = 0.05, \quad \sigma_8 = 0.75, \quad n_s = 0.96,$$

broadly matching the fiducial cosmology used in Maraio, Hall & Taylor (2022) ($h = 0.7$, $\Omega_c = 0.27$, $\Omega_b = 0.045$, $\sigma_8 = 0.75$, $n_s = 0.96$, massless neutrinos). The line-of-sight is discretised into shells of 200 Mpc comoving width out to $z = 3$; a single realisation of the log-normal matter/convergence/shear fields is drawn once per map resolution and reused for every galaxy-density and footprint combination below (density and footprint only affect galaxy sampling, not the underlying fields, so this does not bias the comparison between combinations).

Galaxies are drawn from a single, un-tomographic Smail-type redshift distribution,

$$\frac{dn}{dz} \propto z^2 \exp \left[- \left(\frac{z}{z_*} \right)^{1.5} \right], \quad z_{\text{mode}} = 0.9,$$

normalised to the target surface density $n \in \{1, 10, 20\} \text{ arcmin}^{-2}$, with per-galaxy intrinsic ellipticity dispersion $\sigma_e = 0.3$ drawn from GLASS’s intrinsic-normal ellipticity model. This is analogous to the $\bar{n} = 3$ and $\bar{n} = 30 \text{ arcmin}^{-2}$, single-bin comparison in Maraio, Hall & Taylor (2022) Section 4.3.1, and lets us probe both a shallow, per-tomographic-bin-like density and a deep, all-bins-combined-like density.

Three sky footprints are constructed with GLASS’s combined Galactic+ecliptic visibility cut, using colatitude cuts of $(55^\circ, 125^\circ)$, $(65^\circ, 115^\circ)$ and $(75^\circ, 105^\circ)$ about both the Galactic and ecliptic planes; these give disconnected, moderately-masked and lightly-masked footprints with sky fractions

increasing as the cut narrows the excluded band (Table 1, Figure 1). Two map resolutions are used throughout, $N_{\text{side}} = 64$ and $N_{\text{side}} = 128$ (with $\ell_{\text{max}} = N_{\text{side}}$ in each case), giving $3 \times 3 \times 2 = 18$ independent catalogue/mask combinations in total.

Footprint cut [deg]	$f_{\text{sky}} (N_{\text{side}} = 64)$	$f_{\text{sky}} (N_{\text{side}} = 128)$
(55, 125)	15.26%	15.09%
(65, 115)	30.83%	29.87%
(75, 105)	55.55%	54.22%

Table 1: Sky fraction of the three footprint cuts used, at each resolution (small differences between resolutions are pixelisation effects).

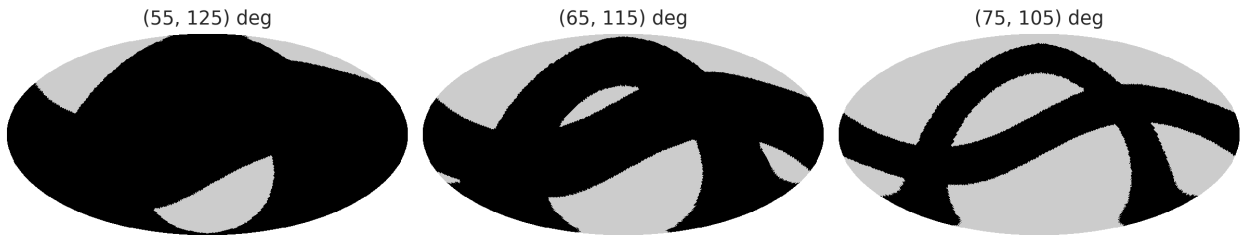


Figure 1: The three footprint masks used in the grid ($N_{\text{side}} = 64$ shown; the $N_{\text{side}} = 128$ masks are visually identical, see Table 1).

For each of the 18 combinations, we run our C++ QML pipeline (Fisher-matrix estimator, finite-difference derivatives, conjugate-gradient linear solves via Eigen, following the method of Maraio, Hall & Taylor (2022) Section 2.3), the NaMaster PCl estimator with its analytic Gaussian covariance, and an independent heracles-based catalogue-to- C_ℓ pipeline (at $2\times$ the QML/PCl map resolution, with its own mixing-matrix deconvolution), and save the resulting EE/BB power-spectrum estimates and the QML/PCl covariance (standard-deviation) ratio per multipole to disk.

3 Log-normal fields and QML validity

The QML estimator’s minimum-variance and unbiasedness properties are formally derived under the assumption of a Gaussian random field. GLASS instead generates log-normal fields, which is a more realistic model of the late-time, non-linear matter and convergence field but formally violates this assumption. We do not repeat that derivation or its robustness tests here; instead we rely directly on Maraio, Hall & Taylor (2022), Section 4.5 (“Non-Gaussian maps”) and their Figure 14, in which the same QML and PCl estimators are applied to both Gaussian and log-normal (via Flask) realisations at $N_{\text{side}} = 256$, and the ratio of PCl to QML standard deviation is found to be “virtually identical” between the two cases. This demonstrates empirically that QML’s variance advantage over PCl persists for log-normal fields, which is the basis for applying our QML pipeline to GLASS’s log-normal-field-based mock catalogues in this document.

4 Results: sky coverage and depth requirements

4.1 $N_{\text{side}} = 64$

Figure 2 shows the ratio of PCI to QML standard deviation, $\sigma_\ell^{\text{PCI}}/\sigma_\ell^{\text{QML}}$, for the EE and BB blocks, across the 3×3 grid of galaxy density (rows) and footprint (columns). A ratio above unity means QML is more precise than PCI at that multipole. The narrowest footprint, $(55, 125)^\circ$ ($f_{\text{sky}} = 15\%$), shows a large QML advantage at low ℓ for both EE and especially BB (up to $\sim 40\times$ lower variance), consistent with Maraio, Hall & Taylor (2022)’s Figure 6, where PCI is sub-optimal by $\sim 20\%$ for EE at $\ell \lesssim 50$ and its BB errors are several times larger than QML’s. As the footprint widens (columns to the right), this advantage shrinks toward unity: once the mask is close to full-sky, PCI and QML become close to equally precise, since PCI’s main disadvantage is E/B mode-mixing and information loss induced by a mask, which matters less as the unmasked fraction grows.

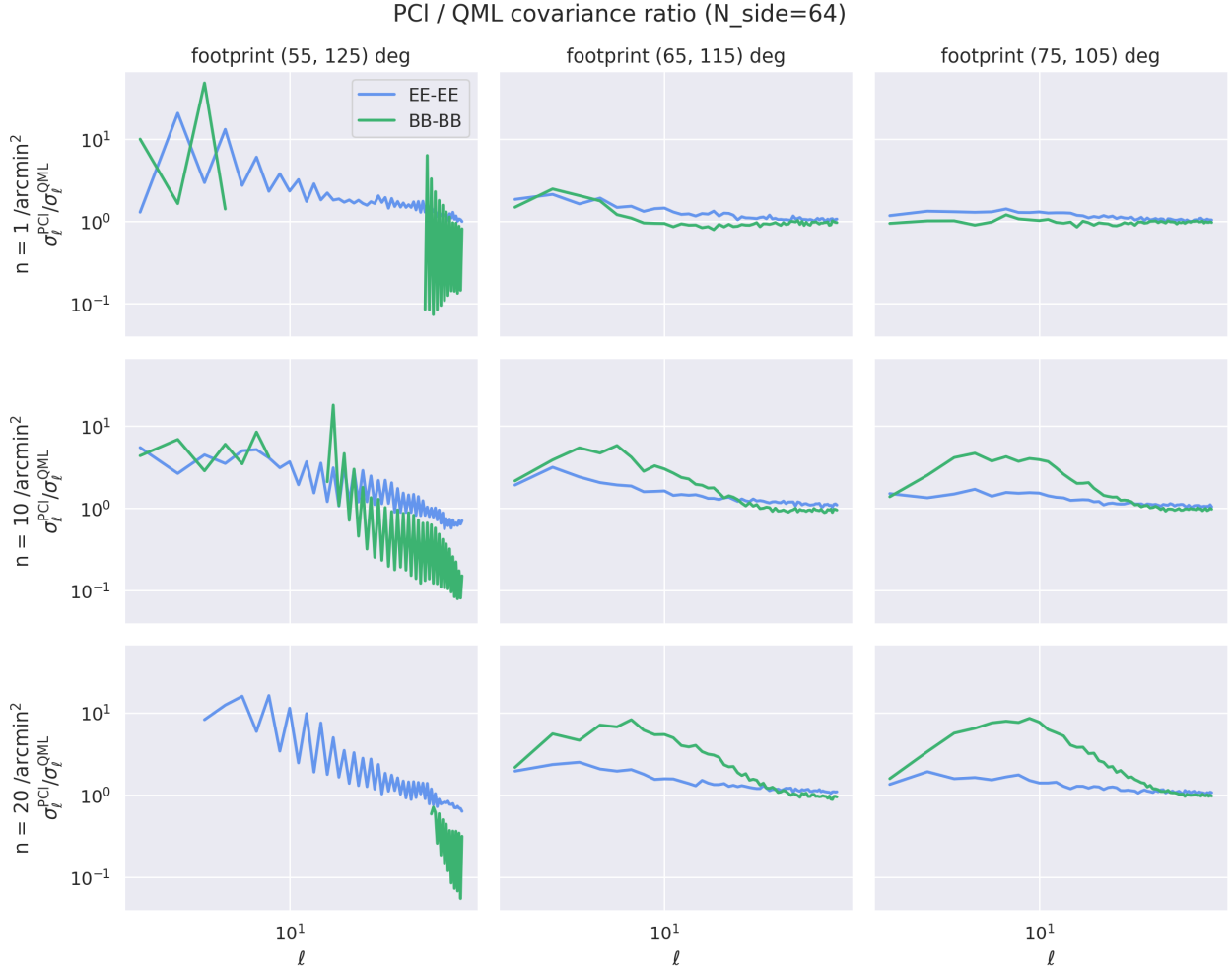


Figure 2: $N_{\text{side}} = 64$: ratio of PCI to QML standard deviation for EE (blue) and BB (green), by galaxy density (rows) and footprint (columns).

Figure 3 shows the recovered EE power spectra themselves. All three estimators track the input theory spectrum well once the footprint is wide enough; with the narrowest footprint the QML estimate becomes visibly noisier (large excursions, including formally negative values which

appear as gaps/spikes on the log-log axes) as the ill-conditioned, highly-masked Fisher matrix becomes harder to invert reliably.

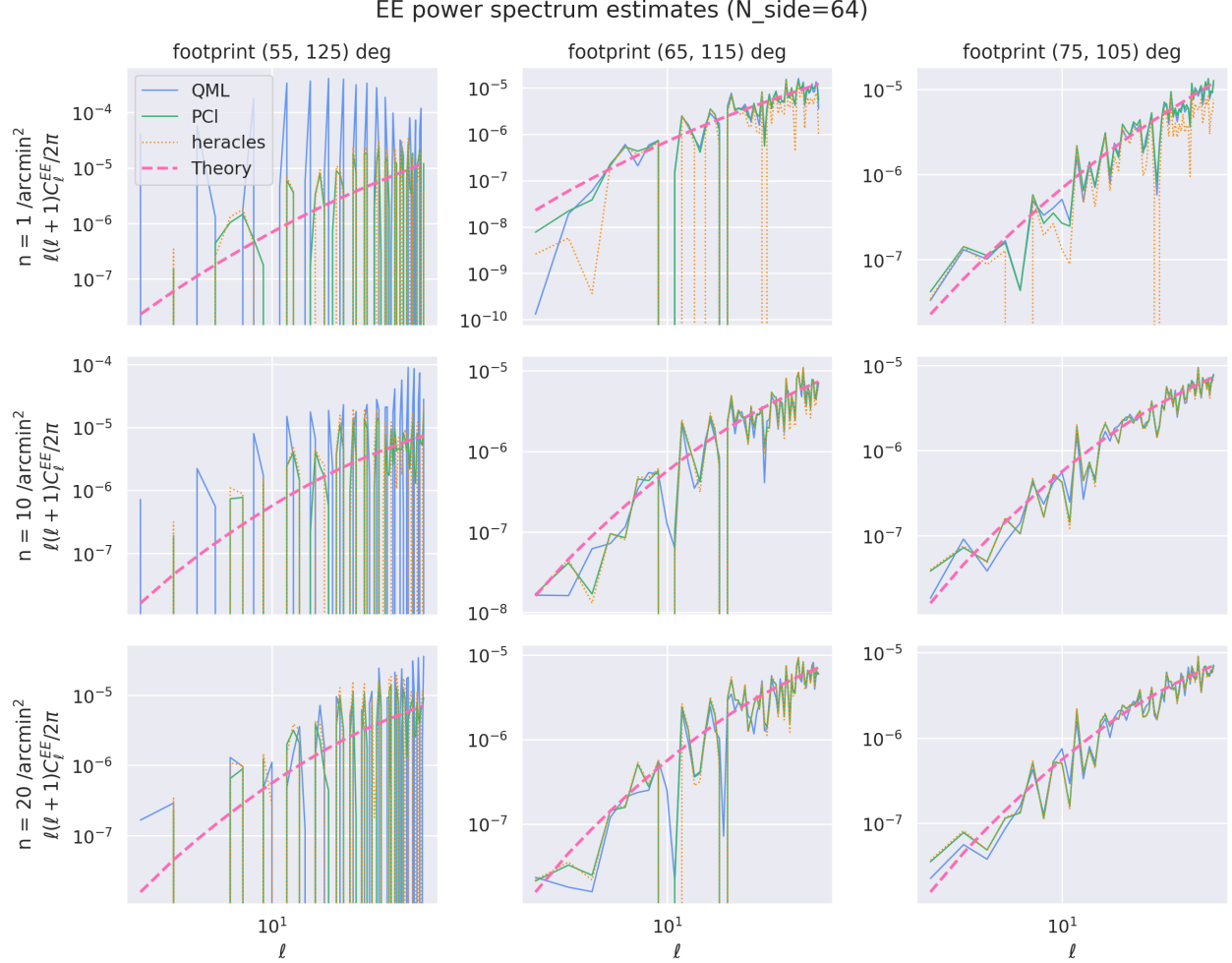


Figure 3: $N_{\text{side}} = 64$: EE power spectrum estimates (QML, PCI, **heracles**, theory), by galaxy density (rows) and footprint (columns).

Figure 4 shows the corresponding BB estimates, which have no true cosmological signal in this simulation and should recover pure shot noise (dashed pink line) if the estimator is behaving well. Table 2 quantifies this as the ratio of the mean recovered QML $|C_{\ell}^{BB}|$ to the theory noise level, both over all ℓ and restricted to $\ell < 30$ where leakage is most severe.

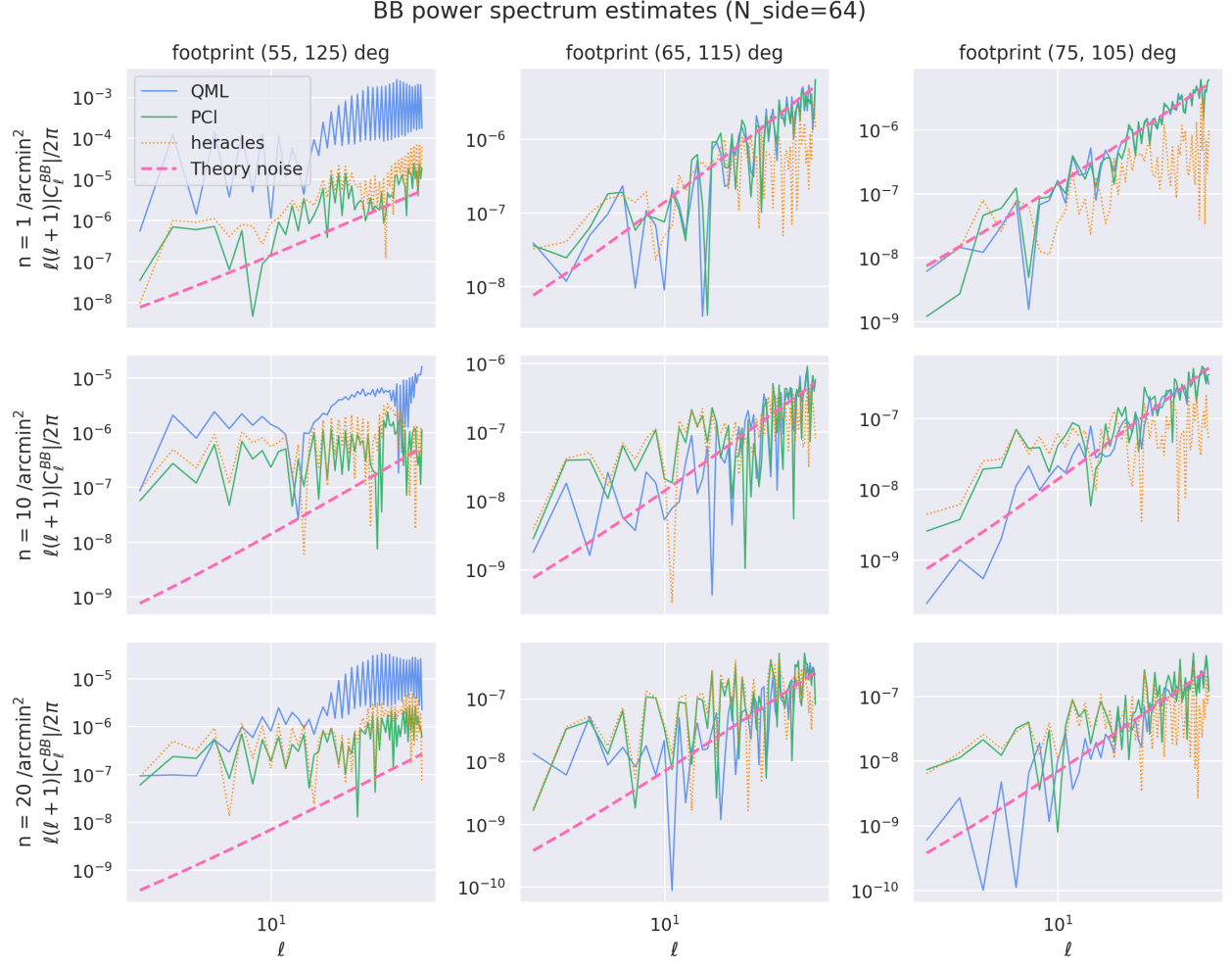


Figure 4: $N_{\text{side}} = 64$: BB power spectrum estimates (QML, PCI, **heracles**, theory noise), by galaxy density (rows) and footprint (columns).

n [arcmin $^{-2}$]	footprint [deg]	BB/noise ($\ell < 30$)	BB/noise (all ℓ)
1	(55, 125)	782.83	566.82
1	(65, 115)	1.22	1.01
1	(75, 105)	0.89	0.94
10	(55, 125)	148.39	79.04
10	(65, 115)	1.76	1.36
10	(75, 105)	1.14	1.05
20	(55, 125)	153.28	134.08
20	(65, 115)	4.66	2.69
20	(75, 105)	1.25	1.13

Table 2: $N_{\text{side}} = 64$: ratio of measured QML $|C_{\ell}^{BB}|$ to the theory shot-noise level. A ratio close to 1 means the recovered B -mode is consistent with pure noise (a “clean” null test).

4.2 $N_{\text{side}} = 128$

The same grid, repeated at twice the resolution, shows the same qualitative behaviour (Figures 6–8, Table 3): a strong QML variance advantage and severe B -mode leakage for the narrowest footprint, and near-unity ratios / noise-consistent B -modes once the footprint is wide. The B -mode/noise excess for the narrow footprint is somewhat less extreme at $N_{\text{side}} = 128$ than at $N_{\text{side}} = 64$ (e.g. $\sim 100\times$ vs. $\sim 800\times$ at $n = 1 \text{ arcmin}^{-2}$, $\ell < 30$), most likely because the extra small-scale information available at higher ℓ_{max} partially dilutes the low- ℓ leakage-dominated average, but the qualitative conclusion is unchanged.

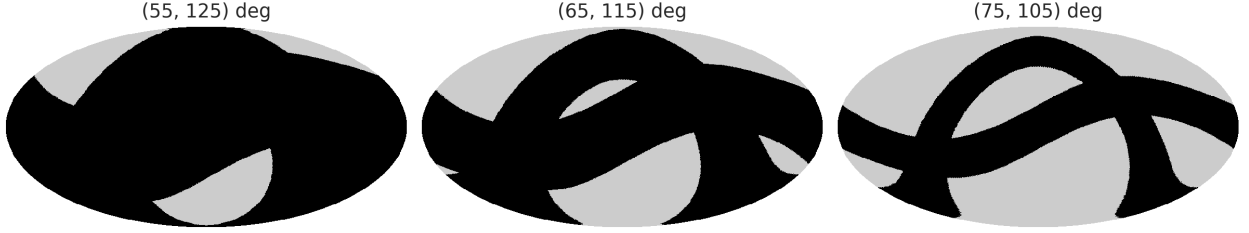


Figure 5: The three footprint masks at $N_{\text{side}} = 128$ (near-identical f_{sky} to $N_{\text{side}} = 64$, Table 1).

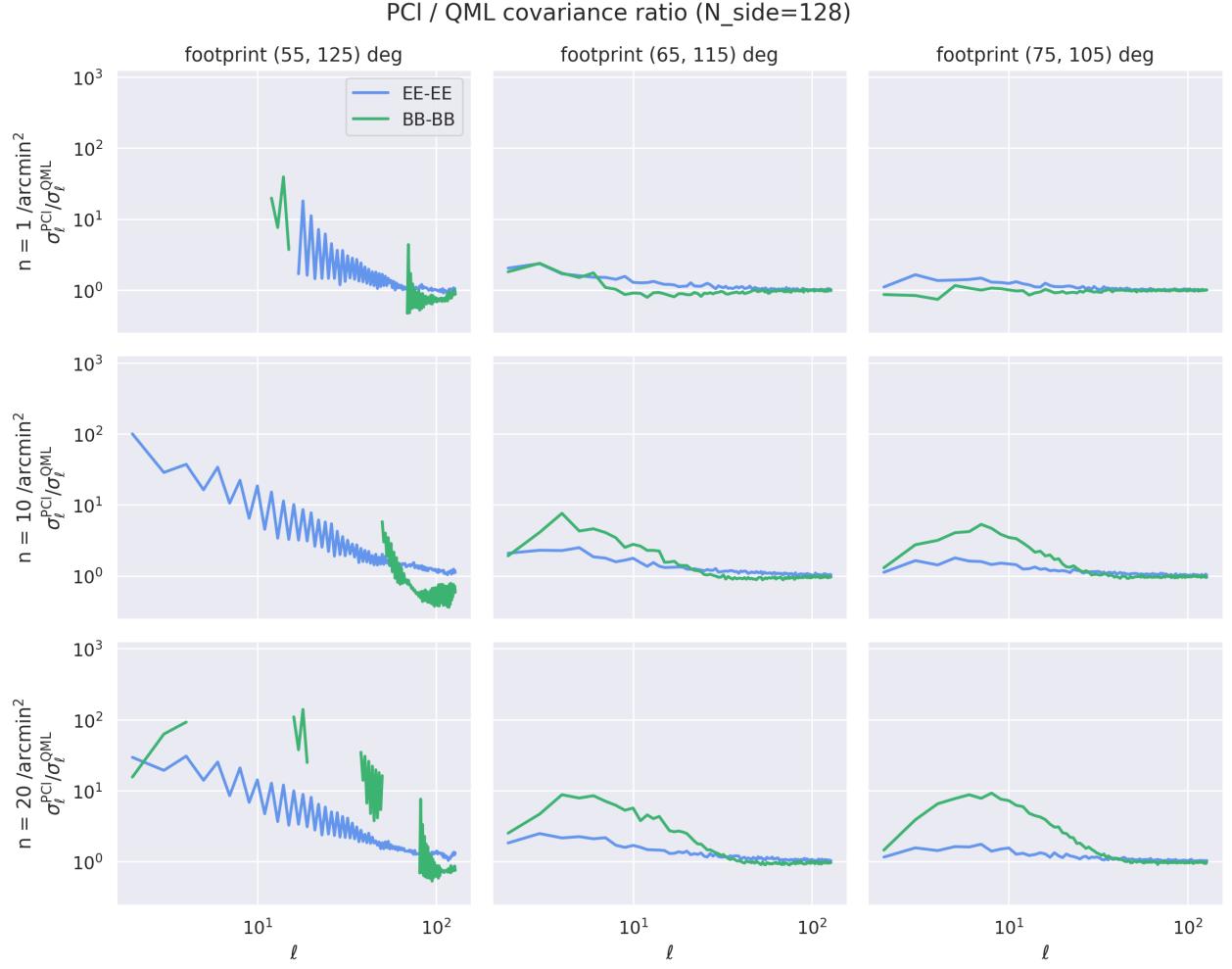


Figure 6: $N_{\text{side}} = 128$: ratio of PCI to QML standard deviation for EE (blue) and BB (green), by galaxy density (rows) and footprint (columns).

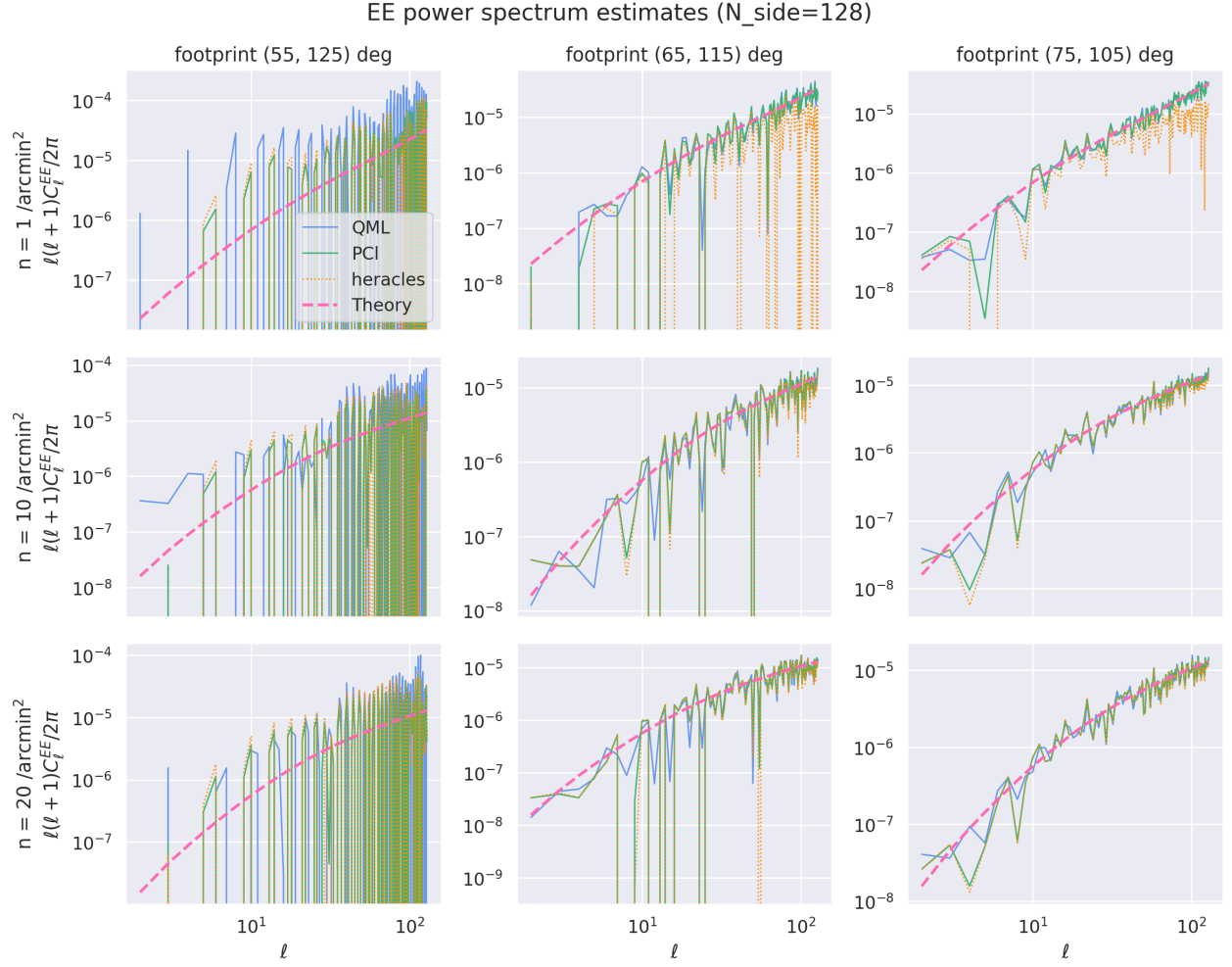


Figure 7: $N_{\text{side}} = 128$: EE power spectrum estimates, by galaxy density (rows) and footprint (columns).

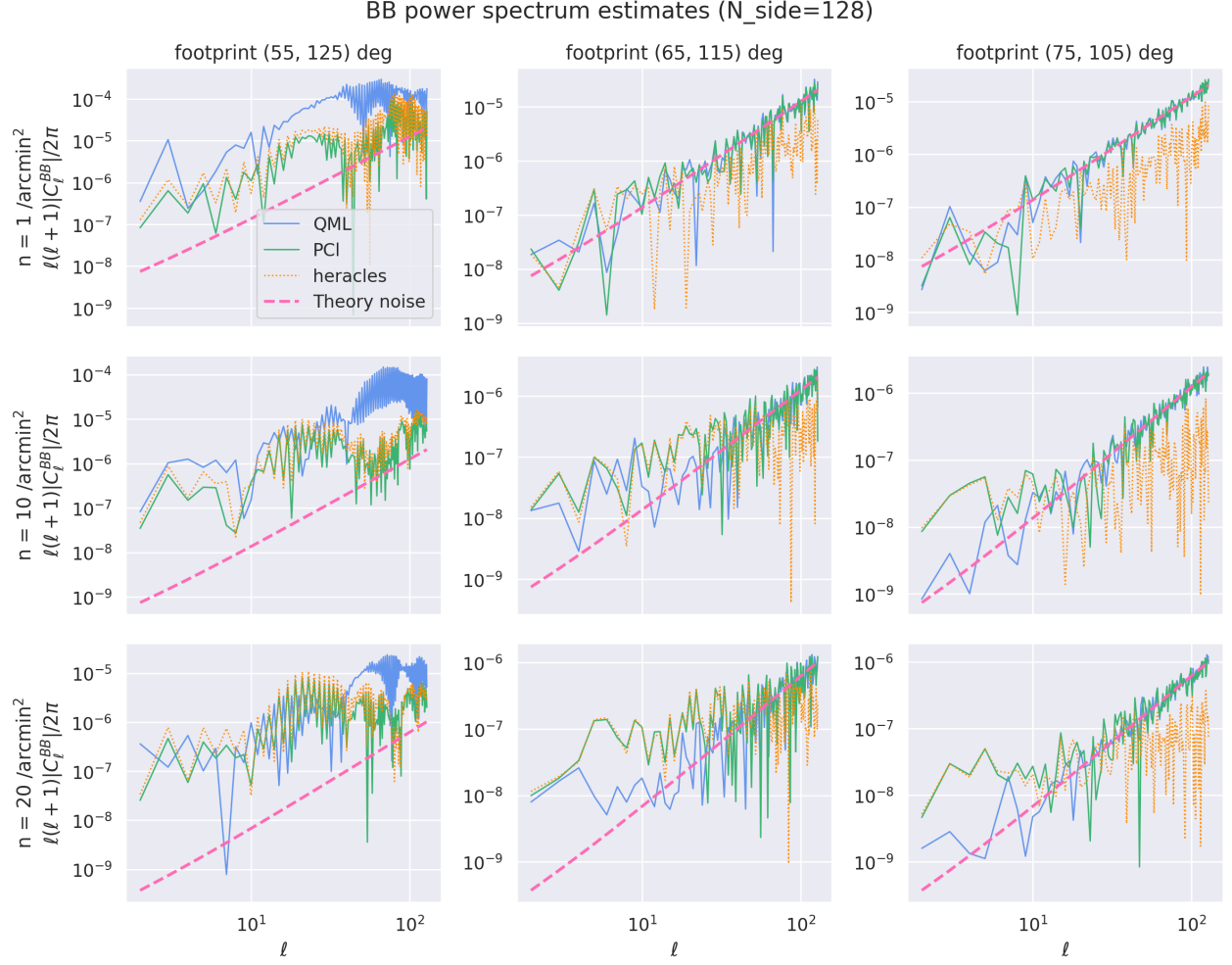


Figure 8: $N_{\text{side}} = 128$: BB power spectrum estimates, by galaxy density (rows) and footprint (columns).

n [arcmin $^{-2}$]	footprint [deg]	BB/noise ($\ell < 30$)	BB/noise (all ℓ)
1	(55, 125)	97.28	42.66
1	(65, 115)	1.45	1.07
1	(75, 105)	1.19	1.04
10	(55, 125)	129.97	86.68
10	(65, 115)	4.19	1.70
10	(75, 105)	1.33	1.08
20	(55, 125)	119.13	50.90
20	(65, 115)	3.93	1.68
20	(75, 105)	1.45	1.10

Table 3: $N_{\text{side}} = 128$: ratio of measured QML $|C_{\ell}^{BB}|$ to the theory shot-noise level.

4.3 Overall finding

Across both resolutions, the grid shows clearly that reproducing Maraio, Hall & Taylor (2022)’s central result – that QML’s noise is meaningfully below PCI’s, particularly for B -modes – requires *both* a sufficiently wide, well-connected sky footprint *and* a sufficiently high galaxy number density:

- **Coverage controls leakage.** With the narrowest, most fragmented footprint ($f_{\text{sky}} \approx 15\%$), the recovered B -mode exceeds the noise floor by one to several orders of magnitude at *every* density tested – mask-induced E -to- B leakage dominates regardless of how many galaxies are used, and the QML/PCI variance ratio, while formally large, is not by itself evidence of a trustworthy measurement.
- **Density controls precision.** With the widest footprint ($f_{\text{sky}} \approx 55\%$), the B -mode stays close to the noise floor at every density, but the absolute size of that noise floor – and hence how informative a “consistent with noise” result is – only becomes small at the highest density tested ($n = 20 \text{ arcmin}^{-2}$).
- Only the combination of wide coverage *and* high density gives both a low absolute B -mode amplitude and a precise, well-constrained measurement of it, matching the regime in which Maraio, Hall & Taylor (2022) demonstrate QML’s advantage over PCI.

5 The real TR1 catalogue

We repeat the sky-coverage and depth diagnostics of Section 4 on the real TR1 Euclid-like shear catalogue, using its effective-coverage HEALPix map degraded to $N_{\text{side}} = 256$ and its per-galaxy `SHE_WEIGHT/TOM_BIN_ID` columns.

5.1 Sky coverage

Figure 9 shows the TR1 effective-coverage mask. Its sky fraction is

$$f_{\text{sky}}^{\text{TR1}} = 0.371\%,$$

roughly $40\text{--}150\times$ smaller than even the narrowest, worst-case footprint used in our grid ($f_{\text{sky}} \approx 15\%$, Table 1), which itself already produced severe B -mode leakage that overwhelmed the shot-noise floor at every density tested (Tables 2–3). By direct extrapolation of that trend, we would expect a QML (or PCI) analysis of the current TR1 footprint to be dominated by mask-induced leakage rather than by the underlying signal or shot noise – i.e. the sky coverage of the current TR1 dataset is too small to support a clean QML null test or precision power-spectrum recovery with this pipeline.

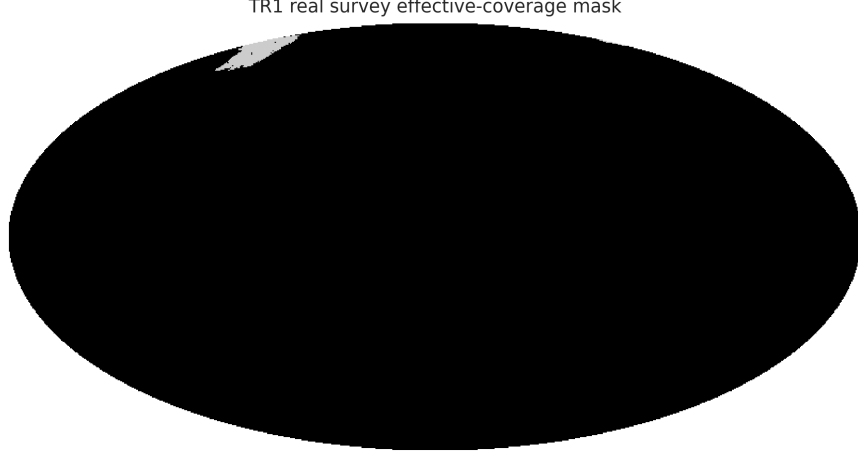


Figure 9: Effective-coverage mask of the real TR1 catalogue, at $N_{\text{side}} = 256$. $f_{\text{sky}} = 0.371\%$.

5.2 Galaxy density

Figure 10 shows the effective galaxy number density ($n_{\text{eff}} = (\sum w)^2 / \sum w^2$, converted to arcmin^{-2} over the unmasked footprint) per tomographic bin and combined across all six bins.

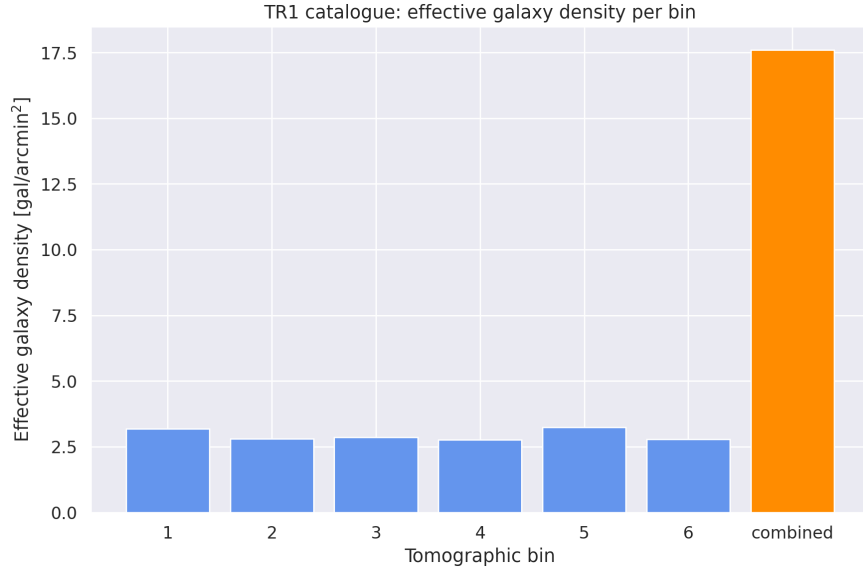


Figure 10: TR1 catalogue effective galaxy density per tomographic bin (blue) and combined across all bins (orange).

The per-bin densities are all in the range $2.77\text{--}3.25 \text{ arcmin}^{-2}$ (combined: 17.6 arcmin^{-2} across all six bins) – almost exactly the $n = 1\text{--}10 \text{ arcmin}^{-2}$ range of our grid, and directly comparable to the $\bar{n} = 3 \text{ arcmin}^{-2}$ per-tomographic-bin case in Maraio, Hall & Taylor (2022) Section 4.3.1. Unlike the coverage deficit above, this depth is *not* the limiting factor: our grid shows that a density of a few galaxies per square arcminute is perfectly workable for QML *provided the footprint is wide enough* (Tables 2–3, $(65, 115)^\circ$ and $(75, 105)^\circ$ columns, $n = 1$ and $n = 10 \text{ arcmin}^{-2}$ rows give BB/noise (all ℓ) ratios between 0.94 and 1.70).

5.3 Feasibility of a per-bin analysis

Combining the two findings above: the bottleneck for applying this QML pipeline to a TR1-like survey on a per-tomographic-bin basis is sky coverage, not galaxy density. Our $N_{\text{side}} = 128$, $n = 1 \text{ arcmin}^{-2}$ grid point is a reasonable proxy for a single TR1-like tomographic bin at higher resolution, and it already achieves a noise-consistent B -mode (BB/noise within $\sim 45\%$ of unity, including at $\ell < 30$) once $f_{\text{sky}} \gtrsim 30\%$ (Table 3). This suggests that, once the observed footprint of a TR1-like survey is enlarged from its current 0.371% to the tens-of-percent level, a per-bin (2–3 arcmin^{-2}) QML analysis at $N_{\text{side}} = 128$ should be achievable with the precision and leakage control demonstrated here, without needing to combine tomographic bins to boost the density further.

6 Conclusion

Using GLASS log-normal mock weak-lensing catalogues, whose validity as an input to the QML estimator is supported by the Gaussian/log-normal comparison in Maraio, Hall & Taylor (2022) Section 4.5, we reproduce that paper’s central finding that QML recovers a lower-variance, leakage-free B -mode null test than PCl, but only once both sky coverage and galaxy density are adequate. Applying the same diagnostics to the real TR1 catalogue shows its current sky coverage ($f_{\text{sky}} = 0.371\%$) is far too small for a reliable QML (or PCl) analysis, while its per-bin galaxy density ($\approx 2.8\text{--}3.2 \text{ arcmin}^{-2}$) already lies in a workable regime. A per-tomographic-bin QML power-spectrum analysis of a TR1-like survey therefore looks feasible at $N_{\text{side}} = 128$, contingent mainly on enlarging the observed footprint rather than on combining tomographic bins for extra depth.

References

A. Maraio, A. Hall, A. N. Taylor, *Testing Quadratic Maximum Likelihood estimators for forthcoming Stage-IV weak lensing surveys*, MNRAS **000**, 1–14 (2022), arXiv:2207.10412.